Prior choice

Outline

Topics

- Strategies for constructing priors. 如何choose prior.
- Situations where choosing the prior matters. 什么时候 prior effect posterior significantly?
- Informative vs non-informative priors.

Rationale

Selection of priors is a necessary step in the construction of Bayesian models. Difficulties that can be encountered in prior choice motivate hierarchical models, introduced this week.

Example

Consider the problem of selecting a prior for a parameter defined on $[0, 1]$, e.g. p in the basic rocket launch success probability example (iid, continuous):

$$p \sim ??? \quad \text{prior.}$$
$$y|p \sim \mathrm{Binom}(n, p). \quad \text{likelihood}$$

General approach

Often, the choice of prior is approached in two stages

- First, pick a distribution **family** 计算方便.
  - in our example: let us pick say $\mathrm{Beta}(\cdot, \cdot)$,
    - a common starting point for distribution with support $[0, 1]$
    - but not the only choice, e.g., an alternative is the Kumaraswamy distribution.
- Then, pick one member of this family
  - e.g. Beta$(1, 2)$, but how to pick these "magic" numbers (called hyper-parameters)?

When does the choice of prior matter?

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- When **data is large**, posterior tends to be **less sensitive** to specification of the prior
  - Theoretical reason: the "Bayesian central limit theorem" (Bernstein von-Mises theorem).

- Not always true (e.g. partially identifiable model).

② When **data is small**, posterior tends to be **more sensitive** to specification of the prior
  - Extreme example:
    - consider a rocket maiden flight,
    - i.e. there is no observation available yet,
    - **Question:** what will be the posterior? ⇒ post. = prior.

Strategies for constructing priors

① "Informative priors": use expert knowledge to determine the prior ("prior elicitation")
  - e.g.: when we build rockets, there is a lot of quality control, so it would be surprising to see very low values for the success probability parameter p.
  - Many prior elicitation techniques developed, see Petrus et al, 2021 for a recent review,
  - but the state of that literature not completely satisfactory.

② "Non-informative priors": use properties of the likelihood to determine prior
  - more advanced, see Robert, Sections 3.5,
  - not automated, case-by-case mathematical derivation often intractable.
  - Automating this into PPLs is an open problem.

③ **Today:** side-step these issues thanks to **Hierarchical models**.
  - We will not remove the need for prior choice, instead we will decrease sensitivity of the posterior to these choices.

Example, continued

Suppose we picked a Beta, so we now have to pick a member of the Beta family.

$$p \sim \text{Beta}(\overset{\alpha}{?}, \overset{\beta}{?})$$
$$y|p \sim \text{Binom}(n, p).$$

Again the numbers to fill in the "?" are known as **hyper-parameters** → α, β, i.e., an hyper-parameter is a parameter of a prior.

Expert elicitation:

- find a rocket expert,
- ask the expert what they think is a reasonable range of values for p, e.g. upper and lower quartiles Q_1, Q_3.
- Use numerical method to fit α, β such that a Beta(α, β) matches the reported quartiles Q_1, Q_3. ⇒ 反推 Beta

$F_{\text{Beta}(\alpha,\beta)}(Q_1) = 0.25$, $F_{\text{Beta}(\alpha,\beta)}(Q_3) = 0.75$.

Graphical models

Outline

Topics

- Directed graphical models.

Rationale

Graphical models help understanding complex models such as hierarchical ones

Directed graphical model

A graphical model is a discrete graph showing the dependencies involved in a forward sampling description of a model. Specifically:

Definition: a directed graphical model is a discrete graph such that:

- the set of vertices is a set of random variables,
- there is an edge $X_1 \to X_2$ if the forward sampling step producing X_2 requires knowing X_1.

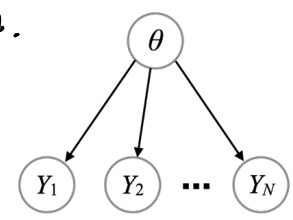
Examples

Exchangeable observations

① y_i iid
② y_i 无 dependence to each other
③ 所有 y_i depend on θ.

Many simples models can be written as:

$$\theta \sim \text{SomePrior}$$
$$y_i|\theta \sim \text{SomeLikelihood}(\theta).$$



Most of the models we have encountered so far have this form.

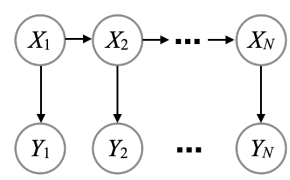
unknown x

只依赖前一时刻

Hidden Markov model

Time series are often modelled as follows:

- there is an unobserved state x_t changing at each time step $t \in \{1, 2, \ldots\}$
  - e.g. true location of a plane,
- we have noisy observations y_t from this true position,
  - e.g. information from a radar.



Simplest HMM example:

$$x_i|x_{i-1} \sim \mathcal{N}(x_{i-1}, \sigma_1)$$
$$y_i|x_i \sim \mathcal{N}(x_i, \sigma_2),$$

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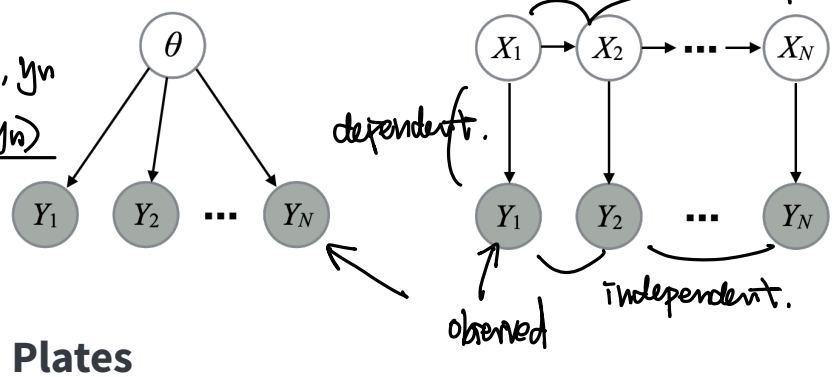
where σ_1 controls the amount of variability in the dynamics, and σ_2, in the measurement.

Conditioning

Shaded nodes in a directed graphical models indicate what we condition on (i.e. what we observe).

$\theta \to y_1, \ldots, y_n$

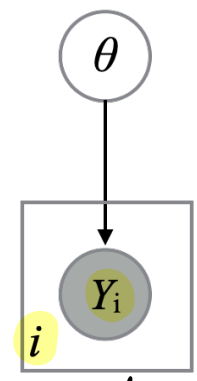
$p(\theta | y_1, \ldots, y_n)$



$p(x_{1:n} | y_{1:n})$

Plates

Plate: When there are too many vertices to draw in a graphical model, use a square ("plate") to indicate a group of nodes that are repeated several times. In other words, a graphical "for loop."



$y_i | \theta \overset{iid}{\sim}$ some likelihood

Further pointers on graphical models

Directed graphical models are also useful to identify conditional independence relationships (recall, A is conditionally independent of B given C if $\Pr(A|C)\Pr(B|C) = \Pr(A \cap B|C)$).

If you are curious, watch this video to see how this is done, or read the original paper.

Intro to hierarchical models

Outline

Topics

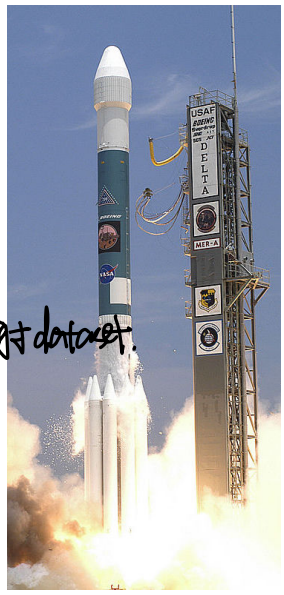
- Hierarchical models.
- Intuition why they decrease prior sensitivity.

Rationale

→ 在N-dataset下, decrease prior sensitivity.

Hierarchical models, a crowning achievement of Bayesian inference, can be used to decrease prior sensitivity even when the dataset of interest is of limited size.

Example



Example: predicting the probability of failure of the next Delta 7925H launch (so far, we observed this type of rocket has been launched 3 times, with 0 failed launches). → small dataset. ⇒ prior = post.

Key idea: use "**side data**" to inform the prior…

其他型号的发射dataset.

- For example: success/failure launch data from other **other types** of rockets.
- Can we use the following rocket launch dataset to inform prediction for a single rocket type of interest?

```
suppressPackageStartupMessages(require("dplyr"))
df = read.csv("../data/failure_counts.csv")
rmarkdown::paged_table(df)
```

LV.Type	numberOfLaunches	numl
<chr>	<int>	
Aerobee	1	
Angara A5	1	
Antares 110	2	
Antares 120	2	
Antares 130	1	
Antares 230	1	
Ariane 1	11	
Ariane 2	6	
Ariane 3	11	
Ariane 40	7	

1-10 of 367 rows Previous **1** 2 3 4 5 6 ... 37 Next

How to use "side data"? Two suboptimal approaches

For pedagogy, we will first cover two simple heuristics to incorporate "side data" and describe their limitations.

Then we will cover hierarchical models in the next section, showing how they address the limitations of simpler methods.

First try (do not use this one!)

①
- Merge all the data into one type? 合并数据 XXX
- I.e. just sum the columns in the data:

```
sum(df$numberOfLaunches)
```

```
[1] 5667
```

```
sum(df$numberOfFailures)
```

```
[1] 431
```

Why merging everything is a bad idea? 假设所有rocket 1个失败 prob.

Towards an improved way to use side data

- Background: "mean–pseudo-sample-size" reparameterization of the Beta distribution.
  - A reparametrization is a different labelling of a family such that you can go back and forth between the two labellings.
  - Consider

$$p \sim \text{Beta}(\alpha, \beta) \Rightarrow \alpha = \mu s, \quad \beta = (1-\mu)s$$

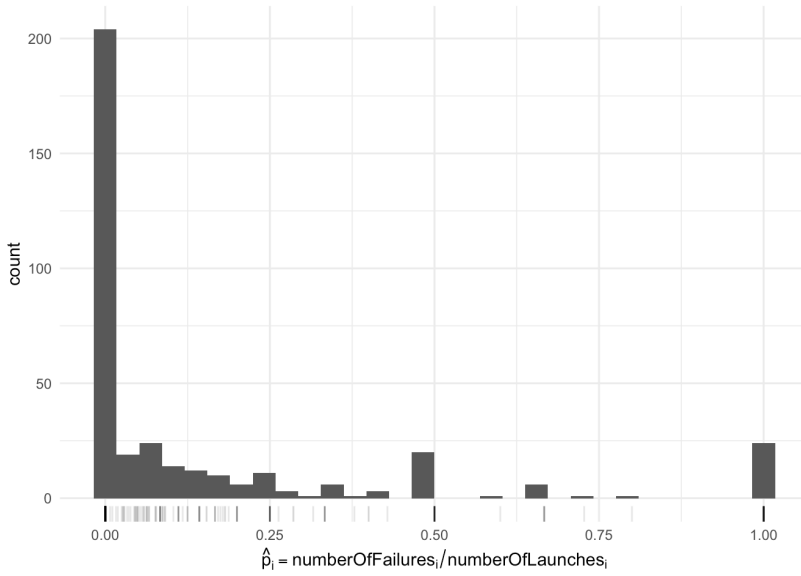
  where $\mu \in (0,1), s > 0$.
- Interpretation:
  - μ: mean of the Beta
  - s: measure of "peakiness" of the density, higher s corresponds to more peaked; roughly, $s \approx$ number of data points that would make the posterior peaked like that.
- Why we did this reparameterization?
  - $\rightarrow$ it should now be more intuitive how we can use "side data" to inform μ…

② Second try (still suboptimal, but not as bad)

- Estimate a failure probability $\hat{p}_i$ for each type of rocket i.
- Fit a distribution $p(\mu)$ on those $\{\hat{p}_i\}$.
- Use this distribution $p(\mu)$ as the prior on μ?

```
suppressPackageStartupMessages(require("ggplot2")
suppressPackageStartupMessages(require("latex2exp
```

```
ggplot(df, aes(x = numberOfFailures / numberOfLau
  geom_histogram() +
  xlab(TeX("$\\hat{p}_i = \\textrm{numberOfFailur
  geom_rug(alpha = 0.1) +
  theme_minimal()
```



Question: what is the weakness of using a prior fitted on this histogram?

⇒ uncertainty of $\hat{p}_i$ is neglected. (有的在1次, 有的在100次)

③ Hierarchical models: a better way to use side data

Solution: go fully Bayesian!

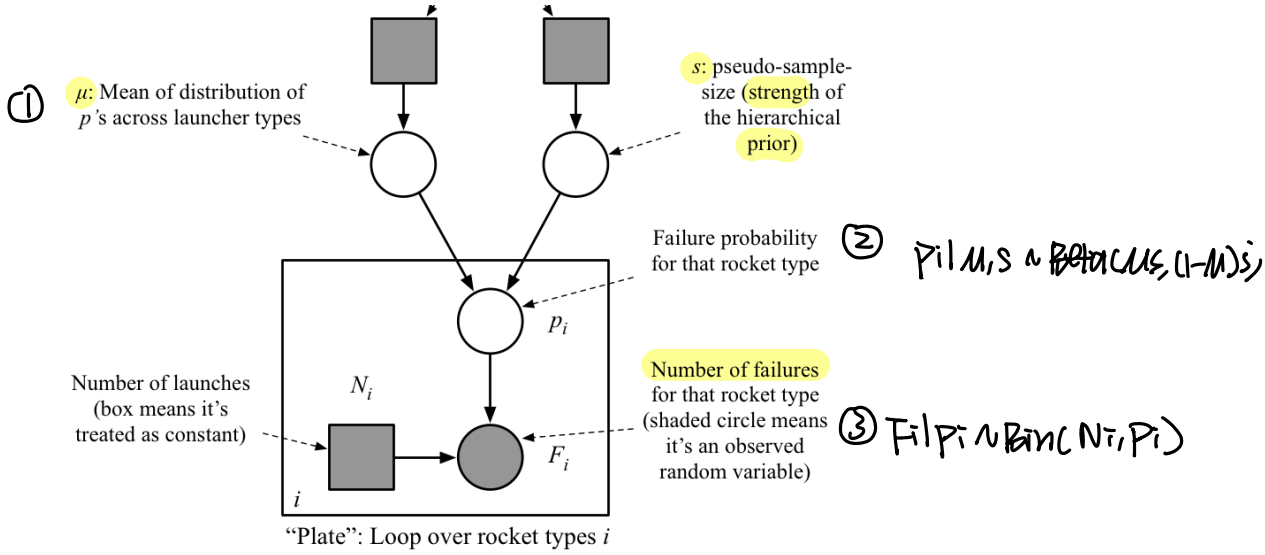
Recall: our Bayesian recipe…

1. Construct a probability model including
   - random variables for what we will measure/observe
   - **random variables for the unknown quantities**
     - those we are interested in ("parameters", "predictions")
     - **others that just help us formulate the problem** ("nuisance", "random effects").
2. Compute the posterior distribution conditionally on the actual data at hand
3. Use the posterior distribution to make a decision

What it means here: we model the launcher type's population parameters (μ, s) as random variables. We are not interested in these other launcher types, but they help us inform inference about the type of rocket we are interested in (i.e., "nuisance" parameters).

Graphical model

New "hyperparameters" introduced by hierarchical construction



Mathematical description

- Share these two "population parameters" across all launch types

prior: $$p_i|\mu, s \sim \text{Beta}(\mu s, (1-\mu)s)$$

- Likelihood same as before: $F_i|p_i \sim \text{Binom}(n_i, p_i)$.
- We still need to put prior on μ and s… → 他们的也要 hyperprior.
  - …but as we'll discuss next, you should expect this prior choice to be less sensitive.
  - Example: $\mu \sim \text{Beta}(1,1) = \text{Unif}(0,1)$, $s \sim \text{Exp}(1/10000)$ rate = 1/mean

PS: why we pick such a small value, $1/10000$?

- Recall the parameter of an exponential is a rate which is 1/mean.
- So a mean of 10000 encodes we put significant mass to values up to $O(10000)$.
- This is sometimes described as a "vague prior". ⇒ 不知道具体表异多大, 但 allowed to be large.

Decrease prior choice sensitivity

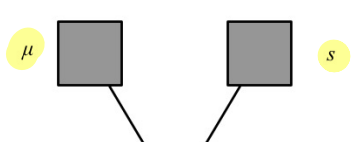
It seems we have introduced new problems as now we again have hyper-parameters, namely those for the priors on μ and s!

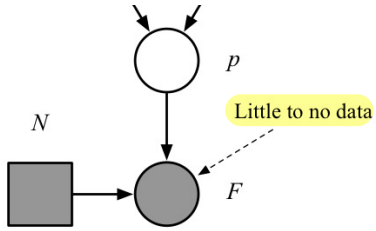
Key point: yes, but now we are less sensitive to these choices! Why?

Heuristic: look at the node n connected to the hyper-parameters…

- If most of n's edges link to hyper-parameters: posterior more sensitive to hyper-parameters.
- If most of n's edges link to random variables connected to data: posterior will probably be less sensitive.

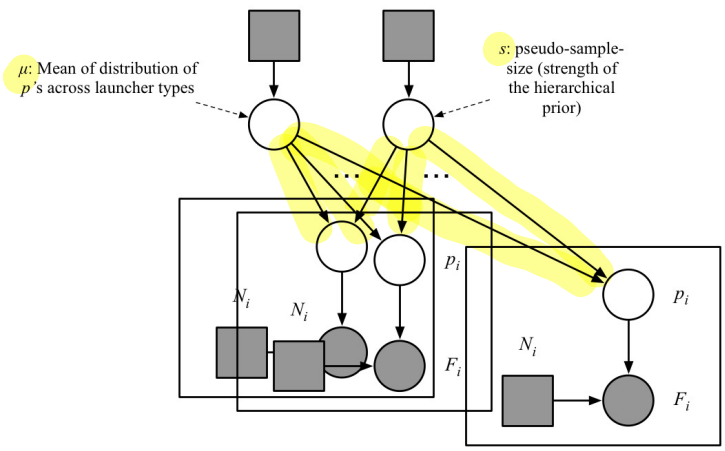
Before going hierarchical: for maiden/early flights we had





Note: p is connected to only a small number of observations.

After going hierarchical:



Note: μ, s is connected to a large population of p_i's, each linked with data.